QUANT

Venn Diagrams

Assignment 03

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Venn Diagram Online Assignment -03

1. In a class of 35 students, 17 have taken Mathematics, 10 have taken Mathematics but not Economics. Find the number of students who have taken both Mathematics and Economics and the number of students who have taken Economics but not Mathematics, if it is given that each student has taken either Mathematics or Economics or both.

1. 7, 15 2. 15, 8 3. 7, 18 4. 5, 12

2. In a town of10,000 families it was found that 40% families buy newspaper A, 20% families buy newspaper B and 10% families buy newspaper C. 5% families buy A and B, 3% buy B and C and 4% buy A and C. If 2% families buy all the three news papers, find the number of families which buy A only.

1. 3250 2. 3300 3. 3100 4. 3375

3. A college awarded 38 medals in Football, 15 in Basketball and 20 to Cricket. If these medals to a total of 58 men and only three men got medals in all the three sports, how many received medals in exactly two of the three sports ?

1. 10 2. 9 3. 11 4. 15

4. In a survey of 100 students, the number of students studying the various languages were found to be: English only 18, English but not Hindi 23, English and German 8, English 26, German i8, German and Hindi 8, no language 24. Find the number of students who were studying Hindi.

1. 12 2. 17 3. 15 4. 18

5. Out of 280 students in class XII of a school 135 play Hockey, 110 play football, 80 play volleyball, 35 of these play hockey and football, 30 play volleyball and hockey, 20 play football and volleyball. Also, each students plays at least one of the three games. How many students play all the three games?

1. 45 2. 40 3. 35 4. 32

6. From 50 students taking examination in Mathematics, Physics and Chemistry, 37 passed Mathematic, 24 Physics and 43 Chemistry. At most 19 passed Mathematics and Physics; at most 29 passed Mathematics and Chemistry and at most 20 passed Physics and Chemistry. If each student has passed in at least one of the subjects, find the largest number of students who could have passed in all the three subjects.

1. 20 2. 22 3. 14 4. 16

7. A school awarded 58 medals for Honest, 20 for Punctuality and 25for Obedience. If these medals were bagged by a total of 78 students and only 5 students got meadals for all the three values, find the number of students who received medals for exactly two of the three values.

1. 15 2. 18 3. 16 4. 10

8. In an university, out of 100 students 15 offered Mathematics only; 12 offered Statistics only; 8 offered Physics only; 40 offered Physics and Mathematics; 20 offered Physics and Statistics; 10 offered Mathematics and Statistics; 65 offered Physics. Find the number of students who offered Mathematics.

1. 60 2. 55 3. 52 4. 62

9. In a survey of 25 students, it was found that 15 had taken Mathematics, 12 had taken sand 11 had taken Chemistry, 5 had taken Mathematics and Chemistry, 9 had taken Mathematics Physics, 4 had taken Physics and Chemistry and 3 had taken all the three subjects. Find the number of that had taken

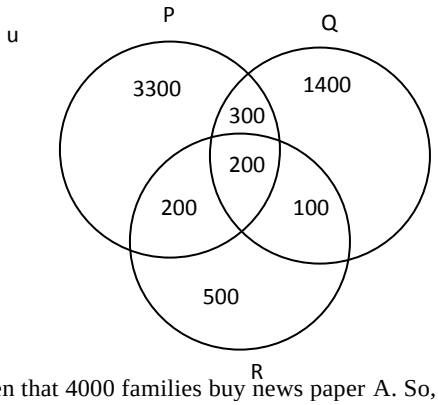
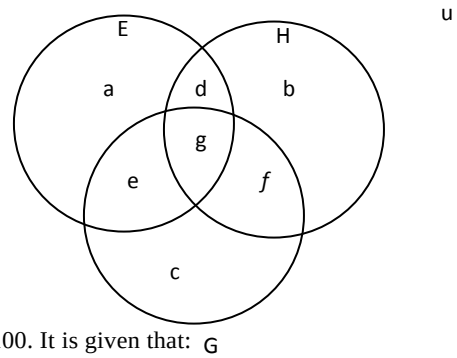
(i) only Chemistry. (ii) only Mathematics.

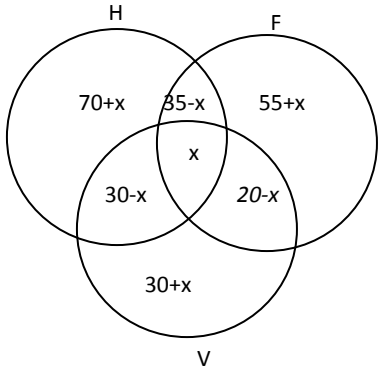
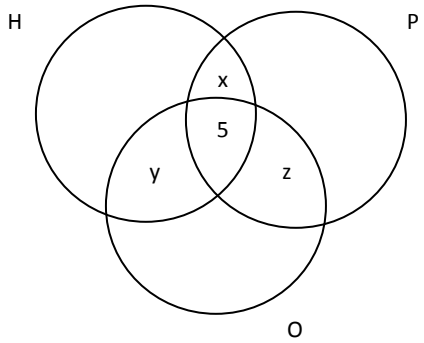
1. 4, 5 2. 5, 5 3. 5, 4 4. 6, 2

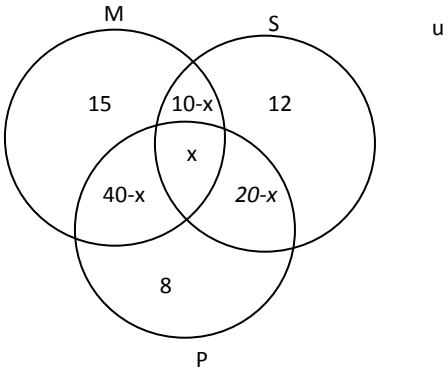
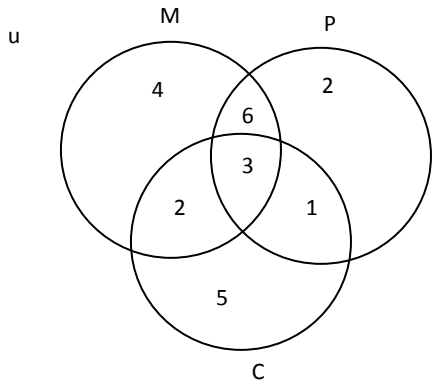
10. In a school there are 20 teachers who teach mathematics or physics. Of these, 12 teach mathematics and 4 teach physics and mathematics. How many teach physics?

1. 10 2. 12 3. 15 4. 16

Answer Key & Explanations

Sr. No.	Key	Explanation
1	3	Let A denote the set of students who have taken Mathematics and B be the set of students who have taken Economics. It is given that n(A∪B) = 35, n (A) = 17 and n (A - B) =10. Now, n(A) = n(A - B) + n(A∩B) ⇒ 17 =10 + n(A∩B) ⇒ n(A∩B) = 7 Thus, 7 students have taken both Mathematics and Economics. Now, n(A∪B) = n(A) + n(B) - n(A∩B) ⇒ 35 = 17 + n(B)-7 ⇒ n(B) Also, n(B) = n(B-A) + n (A∩B) ⇒ 25 = n(B-A) +7 ⇒ n(B-A) = 18. Thus, 18 students have taken Economics but not Mathematics.
2	2	It is given that n (P ∩Q∩ R) = 200 and n (P∩Q) = 500. So, the number of families buying newspaper A and B only is 500-200 = 300. Similarly, the number of families buying news papers B and C only is 300 - 200 = 100 and news papers C and A only is 400 - 200 = 200.  It is given that 4000 families buy news paper A. So, the number of families buying news paper A only = 4000 - (300 + 200 + 200) = 3300.
3	2	Let F denote the set of men who received medals in Football, B the set of men who raved medals in Basketball and C the set of men who received medals in Cricket. It is given that n(F) = 38, n(B) = 15, n(C) = 20, n(F∪B∪C) = 58 and n(F∪B∪C) = 3. n (F∪B∪C) = n(F) + n(B) + n(C) - n(F ∩B) - n(B∩C) - n (F∪C) + n(F∪B∪C) 58 = 38 + 15 + 20 - n (F∩B) - n (B∩C) - n (F∩C) + 3 ⇒ n (F ∩B) + n(B∩C) + n(F∩C) = 76-58 = 18 Number of men who received medals in exactly two of the three sports = n(F∩B) + n(B∩C) + n(F∩C) - 3n (F∩B∩C) = 18 - 3 × 3 = 9 Thus, 9 men received medals in exactly two of the three sports.
4	4	Let E, H and G be the sets of students studying English, Hindi and German respectively. Let u be the set of students surveyed i.e. the universal set. In the above Venn diagram, let a, b, c, d, e, f and g denote the number of students in the respective regions.  Clearly, n (u) = 100. It is given that: Number of students studying English only = 18 ⇒ a = 18 Number of students studying English but not Hindi = 23 ⇒ a + e = 23 Number of students studying English and German = 8 ⇒ e + g = 8 Number of students studying English = 26 ⇒ a + d + e + g = 26 Number of students studying German =48 ⇒ c + e + g + f = 48 Number of students studying German and Hindi = 8 ⇒ g + f = 8 Thus, we obtain a = 18, a + e = 23, e + g = 8, a + e + g + d = 26, e + g + f + c= 48 and g + f = 8 ⇒ a = 18, e = 5, g = 3, d = 0, *f* = 5, c = 35 It is given that 24 students study no language. Therefore, the number of students who study at least one language is 100 - 24 = 76

		i.e. $n(E \cup H \cup G) = 76$ $\Rightarrow a + b + c + d + e + f + g = 76 \Rightarrow 18 + b + 35 + 0 + 5 + 5 + 3 = 76 \Rightarrow b = 10$ The number of students studying Hindi = $b + d + g + g = 10 + 0 + 3 + 5 = 18$.
5	2	Let H, F and V be the sets of students who play hockey, football and volleyball respectively. Let x be the number of students who play all the three games. It is given that 35 students play hockey and football. So, number of student who play hockey and football only is (35 - x). Similarly, the number of students playing various games are written in the regions representing Figure.  It is given that each student plays at least one of the three games. $\therefore n(H \cup F \cup V) = 280$ $\Rightarrow (70 + x) + (35 - x) + (30 - x) + x + (20 - x) + (55 + x) + (30 + x) = 280$ $\Rightarrow 240 + x = 280 \Rightarrow x = 40$ Hence, 40 students play all the three games.
6	3	Let M, P and C be the sets of students passing in Mathematics, Physics and Chemistry respectively. It is given that $n(M \cup P \cup C) = 50$, $n(M) = 37$, $n(P) = 24$, $n(C) = 43$, $n(M \cap P) \le 19$, $n(M \cap C) \le 29$ and $n(P \cap C) \le 20$ We know that $n(M \cup P \cup C) = n(M) + n(P) + n(C) - n(M \cap P) - n(M \cap C) - n(P \cap C) + n(M \cap P \cap C)$ $\Rightarrow 50 = 37 + 24 + 43 - \{n(M \cap P) + n(M \cap C) + n(P \cap C)\} + n(M \cap P \cap C)$ $\Rightarrow 50 = 104 - \{n(M \cap P) + n(M \cap C) + n(P \cap C)\} + n(M \cap P \cap C)$ $\Rightarrow 54 + n(M \cap P \cap C) = n(M \cap P) + n(M \cap C) + n(P \cap C)$ $\Rightarrow 54 + n(M \cap P \cap C) \le 19 + 29 + 20$ $\Rightarrow n(M \cap P \cap C) \le 14$ $\quad [\because n(M \cap P) \le 19, n(M \cap C) \le 29, n(P \cap C) \le 20]$ Hence, the largest number of students that could have passed in all the three subjects is 14.
7	1	Let H, P and O be the sets of students who bagged medals in Honesty, Punctuality and Obedience respectively. It is given that $n(H) = 58$, $n(P) = 20$, $n(O) = 25$, $n(H \cup P \cup O) = 78$ and $n(H \cap P \cap O) = 5$.  Let x denote the number of students who got medals in Honesty and Punctuality only, y denote the number of students who got medals in Honesty and Obedience only and z denote the number of students who got medals in Punctuality and Obedience only. Now, $n(H \cup P \cup O) = 78$ $\Rightarrow n)H) + n(P) + n(O) - n(H \cap P) - n(P \cap O) - n(H \cap O) + n(H \cap P \cap O) = 78$ $\Rightarrow 58 + 20 + 25 - n(H \cap F) - n(P \cap O) - n(H \cap O) + 5 = 78$ $\Rightarrow (H \cap P) + n(P \cap O) + n(H \cap O) = 30$ Hence, required number of students = $x + y + z = 15$
8	4	Let M, S and P be the sets of students who offered Mathematics, Statistics and Physics respectively. Let x be the number of students who offered all the three subjects. It is given that 10 students offered Mathematics and Statistics. Therefore, number of students who offered Mathematics and Statistics but not Physics is 10 - x. Similarly, number of students in different regions are marked in Figure.

		 It given that 65 students offered Physics. ∴ (40 - x) + x + (20 - x) + 8 = 65 ⇒ 68 -x = 65 ⇒ x = 3 From Figure, we find that the number of students who offered Mathematics =15 + (10 - x) + x + 40- x = 65- x = 65-3 = 62
9	3	It is given that 3 students had taken all three subjects and 9 had taken Mathematics and Physics. So, number of students who had taken Mathematics and Physics but not Chemistry is 6 *m* shown in the Venn diagram. 5 students had taken Mathematics and Chemistry and 3 students **had** taken all the three subjects. So, 2 students had taken Mathematics and Chemistry but not Physics. It is given that 15 students had taken Mathematics. So, number of students who had only Mathematics = 15 - (6 + 3 + 2) = 4.  Similarly, we compute the other values shown in the Venn diagram. It is evident from the Venn diagram that the number of students that had taken (i) Chemistry only is *5*. (ii) only Mathematics is 4.
10	2	n(A ∪ B) = n(A) + n(B) - n (A ∩ B) 20 = 12 + n(B) - 4 20 = 8 + n(B) n(B) = 20 – 8 = 12